



BACHELOR OF ENGINEERING (BE)

Mid–Semester Test

August 2016

FLUIDS & THERMODYNAMICS (Module – Thermodynamics A)

TIME ALLOWED: 1 hour and 15 minutes

MARKS: TOTAL: 46 marks

INSTRUCTIONS: Answer **ALL** questions.

All answers except diagrams and sketches ***must be in ink.***

Write your answers in this answer booklet.

Correcting fluid is not permitted.

Exam conditions apply!

NOTES: Marks allocated to each question are shown in brackets beside each question.

Candidates should show ***all working*** in calculations.

Useful Formulae can be found at the end of this exam paper.

Tables of Thermodynamic Data are supplied.

Question 1 – (5 marks)

Figure 1 shows an unlabelled P-v diagram for a pure substance. Fill in the following information on the diagram:

- a) Phase regions:
 - i) Liquid
 - ii) Solid
 - iii) Vapour
 - iv) Liquid + Solid
 - v) Solid + Vapour
 - vi) Liquid + Vapour.
- b) Important features:
 - i) Critical point
 - ii) Triple Line.
- c) Process Lines: (clearly identify/label each process)
 - i) A constant pressure sublimation process
 - ii) A constant temperature boiling process. (from compressed liquid to superheated vapour)

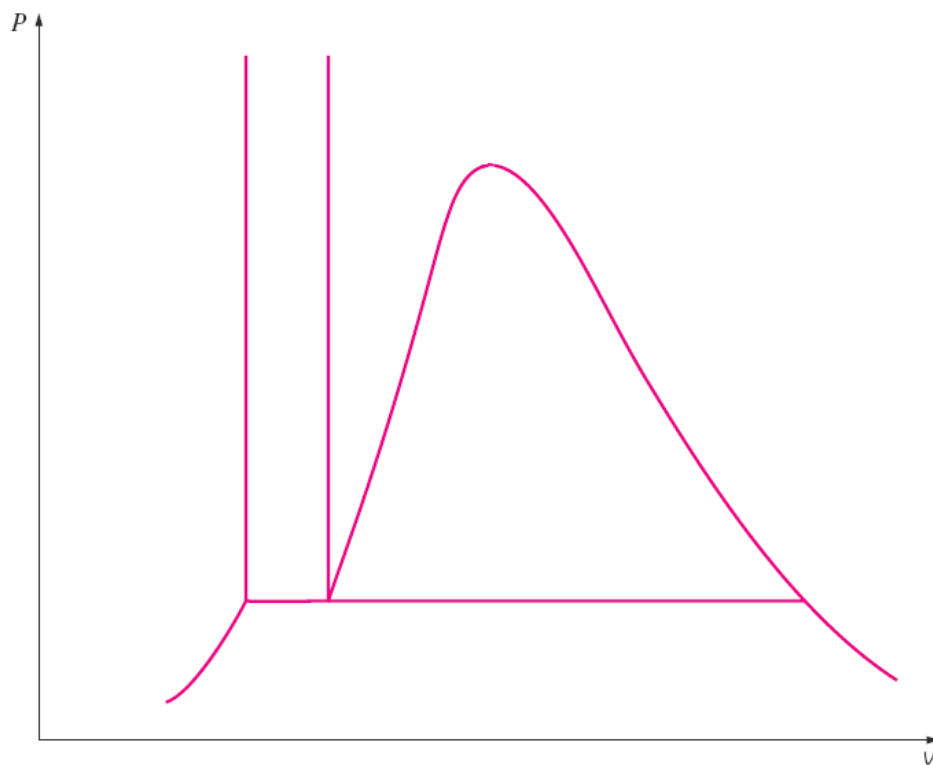


Figure 1

Question 2 – (16 marks)

A piston-cylinder device (Figure 2), whose piston is resting on a set of stops, provides an initial volume that contains 3 kg of refrigerant R-134a at 200 kPa and a quality of $x = 0.2$. Heat is now transferred to the refrigerant until its pressure reaches 400 kPa at which point the piston begins to move. Heat continues to be transferred to the R-134a (the piston maintains a constant pressure of 400 kPa) until the volume increases to twice the initial volume. Determine:

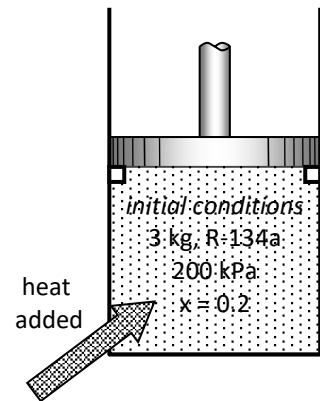


Figure 2

- The initial volume.
- The work done by the expanding R-134a.
- The total heat transferred to the R-134a during this process.

Show the process on a P - v diagram (with respect to saturation lines).

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Question 2 (cont'd)

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Question 3 – (16 marks)

A rigid tank is divided into two equal parts by a partition (Figure 3). Initially, one side of the tank contains 5Kg of water at 200KPa and 25°C, and the other side is evacuated. The partition is then removed and the water expands into entire tank. The water is allowed to exchange heat with its surroundings until the temperature in the tank returns to the initial value of 25°C. Determine:

- The volume of the tank.
- The final pressure.
- The heat transfer for this process.

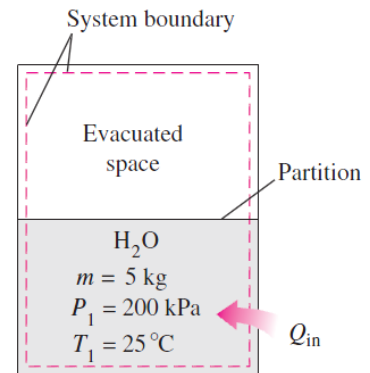


Figure 3

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Question 3 (cont'd)

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Question 4 – (9 marks)

Fill in the missing information in the following table for **water-steam**.

	T (°C)	p (kPa)	u (kJ/kg)	x	Phase description
a.	145		2000		
b.		800	209.33		
c.	270				saturated vapour
d.		30		0.4	
e.		1600	3118.5		
f.	85			0.0	

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Useful Formulae

General & Properties

$$h = u + Pv$$

$$y_x = y_f + xy_{fg}$$

$$x = \frac{m_{\text{sat.vapour}}}{m_{\text{sat.vapour}} + m_{\text{sat.liquid}}}$$

$$P = \rho gz$$

$$v = \frac{V}{m}$$

$$\dot{W}_e = VI$$

Ideal Gas

$$PV = mRT$$

$$PV = NR_u T$$

$$R_u = 8.314 \frac{\text{kJ}}{\text{kmol K}}$$

$$R = \frac{R_u}{M}$$

$$\Delta u = C_v \Delta T$$

$$\Delta h = C_p \Delta T$$

$$C_p = C_v + R$$

$$\gamma = k = \frac{C_p}{C_v}$$

First Law – Closed Systems

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{net energy transfer by heat and work}} = \underbrace{\Delta E_{\text{system}}}_{\text{change in internal, kinetic potential, etc energies}}$$

$$W_{b,\text{out}} = \int_1^2 PdV$$

$$Q - W = \Delta U = m\Delta u$$

$$Q_{\text{net,in}} = W_{\text{net,out}} \quad (\text{closed cycle})$$

First Law – Open Systems

$$\dot{m} = \rho v A = \rho \dot{V} = \frac{\dot{V}}{v}$$

$$v = \text{velocity}$$

Steady Flow Processes

$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}} \quad \dot{Q} - \dot{W} = \underbrace{\sum_{\text{out}} \dot{m} \left(h + \frac{v^2}{2} + gz \right)}_{\text{for each outlet}} - \underbrace{\sum_{\text{in}} \dot{m} \left(h + \frac{v^2}{2} + gz \right)}_{\text{for each inlet}}$$

$$\dot{Q} - \dot{W} = \dot{m} \left[(h_2 - h_1) + \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1) \right]$$

Uniform Flow Processes

$$\sum m_{\text{in}} - \sum m_{\text{out}} = (m_{\text{final}} - m_{\text{initial}})_{\text{system}} = (m_2 - m_1)_{\text{system}}$$

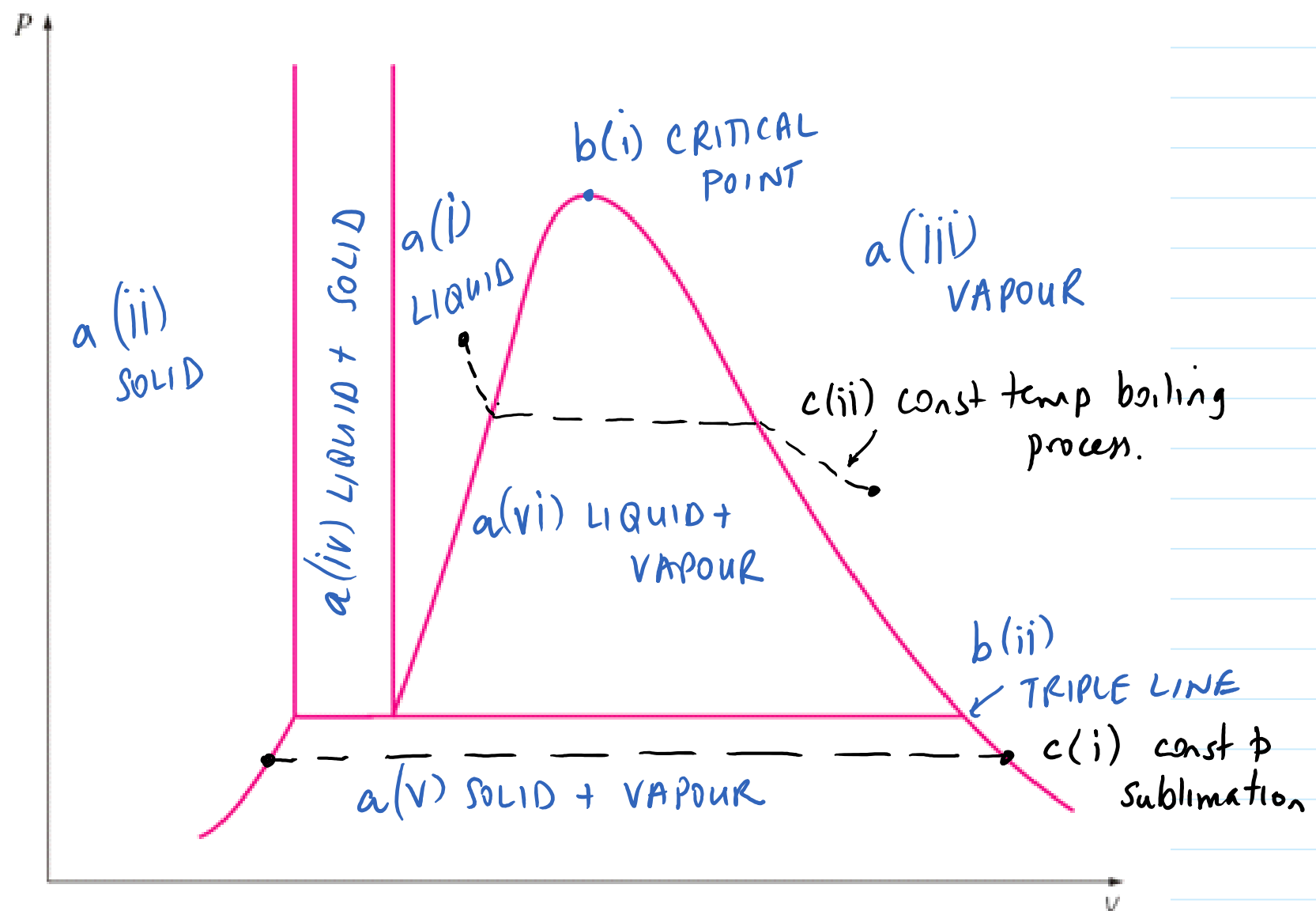
$$Q - W = \sum m_{\text{out}} h_{\text{out}} - \sum m_{\text{in}} h_{\text{in}} + (m_2 u_2 - m_1 u_1)_{\text{system}}$$

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Solutions

Question 1

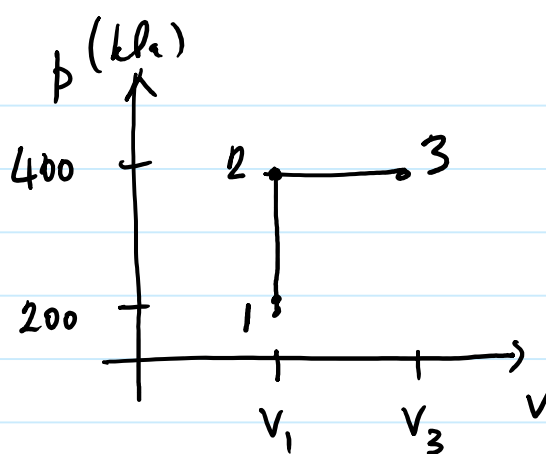
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$\frac{1}{2}$ mark
each
 $\times 10$
 $= 5$

Basic Process:

R134a

not sure where
saturation lines
are yet

$$v_1 = v_2, v_3 = 2v_2$$

(a) State 1: $m = 3 \text{ kg}$, $p_1 = 200 \text{ kPa}$, $x_1 = 0.2$

$$V_1 = m \cdot v_1, \quad v_1 = v_f + x v_{fg} = 0.0007532 + 0.2 \times (0.099951 - 0.0007532)$$

$$\Rightarrow v_1 = 0.020593 \text{ m}^3/\text{kg}$$

$$\Rightarrow V_1 = 3 \times 0.020593 = 0.06178 \text{ m}^3 \quad (\text{initial volume})$$

(b) $W = W_b = \int p \, dV$ area under P-V diagram

$$1 \rightarrow 2 = \text{const. vol process} \Rightarrow W_b = 0$$

$$2 \rightarrow 3 = \text{const. pressure process} \Rightarrow W_b = p_2 (v_3 - v_2)$$

$$\text{Now } V_1 = 0.06178 \text{ m}^3 = V_2$$

$$\text{and } V_3 = 2 \times V_2 = 0.12356 \text{ m}^3$$

$$\Rightarrow W_{13} = 400 \times 10^3 (0.12356 - 0.06178) = 24.711 \text{ kJ} \quad (\text{work out.})$$

(c) First Law - closed system: $Q - W = \Delta U$

$$\Rightarrow Q_{13} - W_{13} = m(u_3 - u_1)$$

need u_1 and u_3

$$200 \text{ kPa} \rightarrow u_1 = u_f + x u_{fg} = 38.26 + 0.2 \times 186.25 = 75.51 \text{ kJ/kg}$$

At state 3 (final state): $p_3 = 400 \text{ kPa}$, $v_3 = 2v_1$

$$\Rightarrow v_3 = 2 \times 0.020593 = 0.041186 \text{ m}^3/\text{kg} \quad (\text{or } V_3/m)$$

From tables at 400 kPa $v_f = 0.0007905 \text{ m}^3/\text{kg}$, $v_g = 0.051266 \text{ m}^3/\text{kg}$

$$v_f < v_3 < v_g \Rightarrow \text{saturated mixture}$$

$$\Rightarrow x_3 = \frac{v_3 - v_f}{v_{fg}} = \frac{0.041186 - 0.0007905}{0.051266 - 0.0007905} = 0.800$$

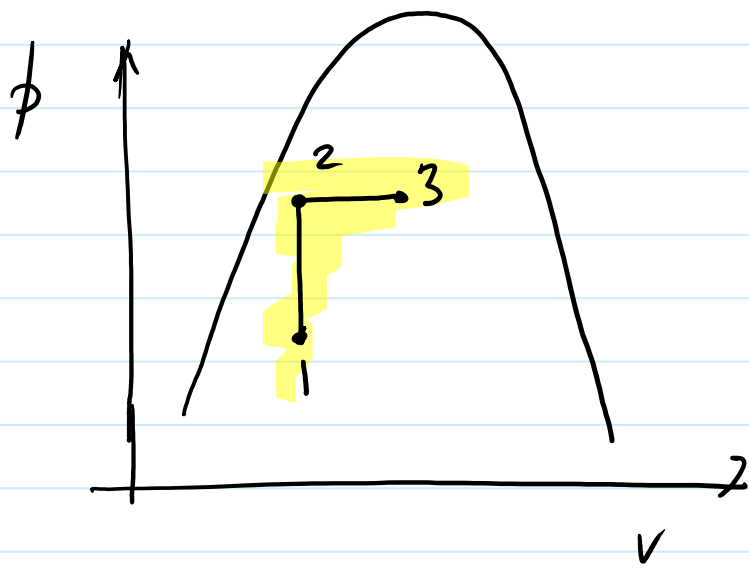
2

$$\Rightarrow u_3 = u_f + x u_{fg} = 63.61 + 0.8 \times 171.49 = 200.8 \frac{\text{kJ}}{\text{kg}}$$

2

$$\begin{aligned} \text{Now } Q_{13} &= W_{13} + m(u_3 - u_1) \\ &= +24.711 + 3(200.802 - 75.51) \\ &= 24.711 + 375.88 \\ &= +400.6 \text{ kJ (heat in)} \end{aligned}$$

2



2

16

(a) At $T_1 = 25^\circ\text{C}$, $p_{\text{sat}} = 3.1698 \text{ kPa} < p_1 \Rightarrow$ compressed liquid

$\Rightarrow v_1 \approx v_f @ 25^\circ\text{C} = 0.001003 \text{ m}^3/\text{kg}$

Now $V_1 = mv_1 = 5 \times 0.001003 = 0.005015 \text{ m}^3$

$V_{\text{tank}} = V_2 = 2 \times V_1 = 2 \times 0.005015 = \underline{\underline{0.01003 \text{ m}^3}}$

(b) $v_2 = \frac{V_2}{m} = \frac{0.01003 \text{ m}^3}{5 \text{ kg}} = 0.002006 \text{ m}^3/\text{kg}$

$T_2 = 25^\circ\text{C} (=T_1) : v_f = 0.001003 \text{ m}^3/\text{kg}, v_g = 43.34 \text{ m}^3/\text{kg}$

$v_f < v_2 < v_g \Rightarrow$ saturated mixture

$\Rightarrow p_2 = p_{\text{sat}} @ 25^\circ\text{C} = \underline{\underline{3.1698 \text{ kPa}}}$

(c) $Q_{12} - \cancel{W_{12}}^0 = \Delta U = m(u_2 - u_1)$

need u_1 and u_2 : $u_1 = u_f @ 25^\circ\text{C} = 104.83 \frac{\text{kJ}}{\text{kg}}$

$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.002006 - 0.001003}{43.34 - 0.001003} = 2.314 \times 10^{-5}$

$\Rightarrow u_2 = u_f + x \cdot u_{fg} = 104.83 + 2.314 \times 10^{-5} \cdot 2304.3$

$\Rightarrow u_2 = 104.88 \text{ kJ/kg}$

$\Rightarrow Q_{12} = 5(104.88 - 104.83) = \underline{\underline{0.267 \text{ kJ}}}$

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Question 4

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	$T, ^\circ\text{C}$	P, kPa	$u, \text{kJ/kg}$	x	Phase description
(a)	145	415.68	2000	0.71	sat. mixture
(b)	50	800	209.33	—	comp. liquid
(c)	270	5503.0	2593.7	1	saturated vapour
(d)	69.09	30	1160.6	0.4	sat. mixture
(e)	500	1600	3120.1	—	superheated vapour
(f)	85	57.868	355.96	0.0	saturated liquid

$\frac{1}{2}$ mark
each
 $\times 18$

$\frac{9}{9}$

(a) $u_f < u < u_g \Rightarrow$ sat. mixture and $p = p_{\text{sat}} = 415.68 \text{ kPa}$

$$x = \frac{u - u_f}{u_{fg}} = \frac{2000 - 610.19}{1944.2} = 0.7148$$

(b) $u < u_f \Rightarrow$ compressed liquid... need to find T where
 $u_f = 209.33 \rightarrow 50^\circ\text{C}$ (Table A4)

(c) sat. vapour $\Rightarrow x = 1$, $u = u_g @ 270^\circ\text{C} = 2593.7 \text{ kJ/kg}$
 and $p = p_{\text{sat}} = 5503 \text{ kPa}$

(d) $x = 0.4 \Rightarrow$ sat. mixture, $T = T_{\text{sat}} @ 30 \text{ kPa} = 69.09^\circ\text{C}$
 $u = u_f + x \cdot u_{fg} = 289.24 + 0.4 \times 2178.5 = 1160.64$

(e) At 1750 kPa $u > u_g \Rightarrow$ superheated at 1600 kPa $\rightarrow$ table A6
 at $p = 1.6 \text{ MPa}$, $u = 3120.1 \text{ kJ/kg}$ when $T = 500^\circ\text{C}$

If used $u = 3118.5 \frac{\text{kJ}}{\text{kg}}$ then $\frac{T - 400}{500 - 400} = \frac{3118.5 - 2950.8}{3120.1 - 2950.8}$

$\Rightarrow T = 400 + 100 \times 0.9905 = 499.1^\circ\text{C}$

(f) $x = 0 \Rightarrow$ saturated liquid, $u = u_f @ 85^\circ\text{C} = 355.96 \frac{\text{kJ}}{\text{kg}}$
 and $p = p_{\text{sat}} = 57.868 \text{ kPa}$